**7** An alternating voltage $V$ varies with time $t$ according to

$$V = 18\cos 40\pi t$$

where $V$ is in V and $t$ is in s.

**(a)** For the alternating voltage:

**(i)** show that the period is 0.050 s

[1]

**(ii)** determine the root-mean-square (r.m.s.) voltage.

r.m.s. voltage = ..... V [1]

**(b)** On Fig. 7.1, sketch the variation of $V$ with $t$ for values of $t$ from $t = 0$ to $t = 100$ ms.

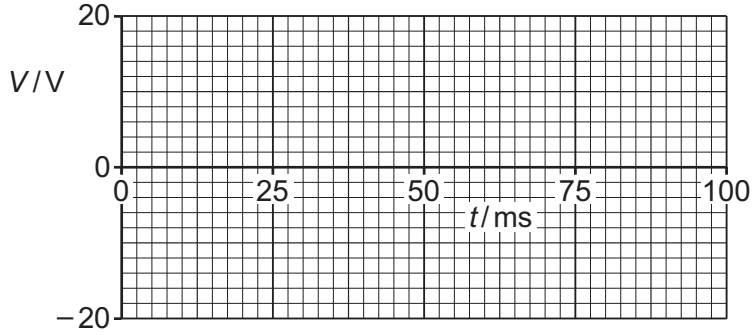



**Fig. 7.1**

[3]

**(c)** The alternating voltage is rectified to produce an output voltage across a load resistor R, as shown in Fig. 7.2.

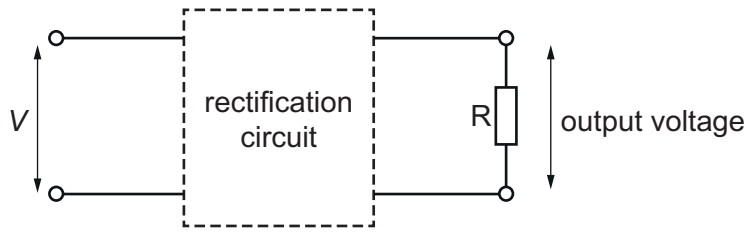



**Fig. 7.2**

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Fig. 7.3 shows the variation with $t$ of the power $P$ in the load resistor.

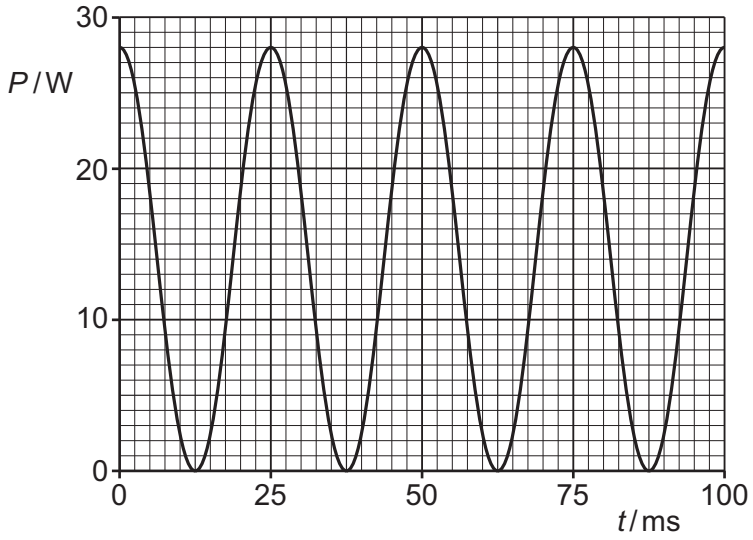



**Fig. 7.3**

State **three** conclusions that can be drawn from Fig. 7.3. The conclusions may be qualitative or quantitative. Use the space for any working.

1 .....

.....

2 .....

.....

3 .....

.....

[3]

[Total: 8]

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